Unit 1

1 Mark Questions with Answers

Introduction to Python

1. **Q:** Who developed Python?
   A: Guido van Rossum.

2. **Q:** In which year was Python released?
   A: 1991.

3. **Q:** Is Python compiled or interpreted?
   A: Interpreted.

4. **Q:** Write one feature of Python.
   A: Simple and easy-to-learn syntax.

5. **Q:** Mention one popular application area of Python.
   A: Web development.

Applications of Python

6. **Q:** Which library is used for Machine Learning in Python?
   A: Scikit-learn / TensorFlow.

7. **Q:** Name one Python framework for web development.
   A: Django or Flask.

8. **Q:** Which library is used for data analysis?
   A: Pandas.

9. **Q:** Name one GUI toolkit in Python.
   A: Tkinter.

10. **Q:** Which Python library is used for scientific computing?
    A: NumPy.

Installation & IDEs

11. **Q:** From which website can you download Python?
    A: www.python.org.

12. **Q:** What does IDE stand for?
    A: Integrated Development Environment.

13. **Q:** Write any two Python IDEs.
A: PyCharm, Jupyter Notebook.

14. **Q:** Which IDE comes by default with Python?
A: IDLE.

15. **Q:** Command to check Python version?
A: `python --version`.

Python Syntax, Indentation & Comments

16. **Q:** What is used to define code blocks in Python?
A: Indentation.

17. **Q:** Which symbol is used for single-line comments?
A: `#`.

18. **Q:** How do you write multi-line comments?
A: Using triple quotes `'''` or `"""`.

19. **Q:** Write a simple if statement in Python.
A:

```python
if True:
    print("Hello")
```

20. **Q:** What error occurs if indentation is missing?
A: IndentationError.

Variables, Data Types & Type Casting

21. **Q:** How do you create a variable in Python?
A: By assigning value directly, e.g., `x = 10`.

22. **Q:** Can a variable hold values of different types in Python?
A: Yes.

23. **Q:** Data type of `3.14` is?
A: float.

24. **Q:** Data type of `"Hello"` is?
A: string.

25. **Q:** What is the output of `float(5)`?
A: 5.0.

Operators

26. **Q:** Which operator is used for exponentiation?
    A: `**`.

27. **Q:** Difference between `==` and `is`?
    A: `==` checks values, `is` checks identity.

28. **Q:** Write one logical operator.
    A: `and`.

29. **Q:** Which operator checks membership?
    A: `in`.

30. **Q:** What does `&` do in Python?
    A: Bitwise AND.

Input/Output Functions

31. **Q:** Which function is used to take input from user?
    A: `input()`.

32. **Q:** Write a print statement.
    A: `print("Hello World")`.

33. **Q:** How to print multiple values in one line?
    A: Using commas, e.g., `print("A", "B")`.

34. **Q:** Which function converts input to integer?
    A: `int()`.

35. **Q:** What is the default separator in `print()`?
    A: Space.

Control Structures

36. **Q:** Write syntax of `if` statement.
    A:

```
if condition:
    statement
```

37. **Q:** Which statement is used to check multiple conditions?
    A: if-elif-else.

38. **Q:** Example of nested if?

A:

```
if x > 0:
    if x % 2 == 0:
        print("Positive Even")
```

39. **Q:** Which statement executes if condition is false?
    A: else.

40. **Q:** Why is elif used?
    A: To check multiple conditions when `if` fails.

Looping

41. **Q:** Which loop is used to iterate over a sequence?
    A: for loop.

42. **Q:** Print numbers fist 10 natural numbers using for loop.
    A:

```
for i in range(1, 11):
    print(i)
```

43. **Q:** Which keyword is used to exit a loop?
    A: break.

44. **Q:** Which keyword skips current iteration?
    A: continue.

45. **Q:** What does `pass` do?
    A: Does nothing (placeholder).

46. **Q:** Can loops be nested in Python?
    A: Yes.

47. **Q:** Print even numbers from 2 to 10 using while loop.
    A:

```
i = 2
while i <= 10:
    print(i)
    i += 2
```

48. **Q:** What happens if condition in while loop never becomes false?
    A: Infinite loop.

49. **Q:** Which statement is used to terminate loop immediately?
    A: break.

50. **Q:** What happens if we use else without if?
A: SyntaxError.

2-Mark Questions with Answers

1. **List any two important features of Python.**
   - Easy to learn and use (simple syntax).
   - Interpreted and dynamically typed.
2. **Give two examples of real-world applications of Python.**
   - Web development (Django, Flask).
   - Data science and machine learning (NumPy, pandas, scikit-learn).
3. **Differentiate between compiler and interpreter in Python installation.**
   - Compiler translates the whole program at once, while **Python uses an interpreter** that executes line by line.
4. **What is the role of an IDE in Python programming?**
   - IDE provides an environment to **write, run, debug, and test Python code** easily (e.g., PyCharm, VS Code).
5. **Write two rules for Python syntax and indentation.**
   - Indentation is mandatory for blocks.
   - Statements do not end with semicolons by default.
6. **Why are comments used in Python? Give an example.**
   - Comments make code readable and explain its purpose.
   - Example: `# This is a single-line comment`
7. **Differentiate between mutable and immutable variables in Python with examples.**
   - Mutable: Can be changed (e.g., `list = [1,2,3]`).
   - Immutable: Cannot be changed (e.g., `tuple = (1,2,3)`).
8. **Write examples of two numeric data types in Python.**
   - Integer → `x = 10` Float → `y = 3.14`
9. **What is type casting? Give one example of implicit and explicit type casting.**
   - Type casting: Converting one data type to another.
   - Implicit: `x = 5; y = 2.5; z = x + y` → result is float.

- Explicit: `int(3.14)` → `3`

10. **Write the difference between relational and logical operators in Python.**

- Relational: Compare values (e.g., `>`, `<`, `==`).
- Logical: Combine conditions (e.g., `and`, `or`, `not`).

11. **What is the use of the membership operator in Python? Give an example.**

- Checks if a value exists in a sequence.
- Example: `"a" in "apple"` → True.

12. **Write the difference between the identity operator `is` and equality operator `==`.**

- `==` checks values → `10 == 10` → True.
- `is` checks memory reference → `[1,2] is [1,2]` → False.

13. **Explain the role of bitwise operators with an example.**

- Perform operations on bits.
- Example: `5 & 3 = 1` (binary AND).

14. **What is the difference between input() and print() functions in Python?**

- `input()` → reads user input.
- `print()` → displays output.

15. **Write a program snippet using if-else to check if a number is even or odd.**

```
n = 4
if n % 2 == 0:
   print("Even")
else:
   print("Odd")
```

16. **What is the difference between if-elif-else and nested if statements?**

- `if-elif-else`: Multiple conditions in sequence.
- Nested if: One `if` inside another.

17. **Write a short program using a for loop to print the first 5 natural numbers.**

```
for i in range(1, 6):
   print(i)
```

18. Write a while loop program to print numbers from 1 to 5.

```
i = 1
while i <= 5:
```

```
    print(i)
    i += 1
```

19. **What is the use of the break statement in Python? Give an example.**
    - Used to exit a loop immediately.

```
for i in range(10):
    if i == 5:
        break
    print(i)
```

20. Differentiate between continue and pass statements with examples.

- **`continue`: Skips current iteration.**

```
for i in range(5):

    if i == 2:

        continue

    print(i)
```

- **`pass`: Does nothing (placeholder).**

```
for i in range(5):

    if i == 2:

        pass

    print(i)
```

5-Mark Questions with Answers

1. Explain the main features of Python with examples.

Answer: Elaborate the points

- **Simple & Easy** → Python has readable syntax.
- **Interpreted** → Executes line by line.
- **Portable** → Same code runs on Windows/Linux/Mac.
- **Dynamically Typed** → No need to declare data type.
- **Extensive Libraries** → Supports NumPy, Pandas, etc.

2. Write a short note on Python applications in real life.

Answer: Elaborate the points
Python is widely used in:

- **Web Development –** Django, Flask.
- **Data Science & AI –** TensorFlow, scikit-learn.
- **Game Development –** Pygame.
- **Automation/Scripting –** Writing automation scripts.
- **Networking & Cybersecurity –** Packet analysis and security testing.

3. Compare compiler and interpreter. Why is Python called an interpreted language?

Answer: Elaborate the points

- **Compiler** translates the whole program into machine code before execution.
- **Interpreter** translates line by line at runtime.
- **Python uses an interpreter** → It executes one line at a time, making debugging easier but execution slower compared to compiled languages like C++.

4. Explain Python variables and data types with examples.

Answer:

- **Variable:** A named storage for values.

  x = 10 # integer

  y = "Hello" # string

- **Data Types in Python:**

i. Numeric (int, float, complex)

ii. Sequence (list, tuple, string)

iii. Set & Dictionary

iv. Boolean

5. Explain type casting in Python with examples.

Answer: Elaborate it

- **Type Casting** → Converting from one data type into another.
- **Implicit Casting:** Python automatically converts types.

  Eg:1 # Integer to float

  x = 5

  y = 2.5

z = x + y # result = 7.5 (float)

Eg 2: Integer to complex

x = 5

y = 6 + 7j

result = x+y # result = 11 + 7j (Complex)

- **Explicit Casting:** Using built-in functions.

  Eg1: float to integer

  num = int(3.7) # result = 3

  Eg2: int to string

  text = str(123) # result = "123"

6. Explain control structures in Python with examples.

Answer:

- **if statement Eg:**

```
x = 10
if x > 5:
   print("Greater")
```

- **if-else statement Eg:**

```
x = 3
if x % 2 == 0:
   print("Even")
else:
   print("Odd")
```

- **if-elif-else Eg:**

```
marks = 75
if marks >= 90:
        print("Grade A")
elif marks >= 60:
        print("Grade B")
else:
        print("Grade C")
```

7. Explain the use of loops in Python with examples of for and while loops.

Answer:

- **for loop:** Iterates over a sequence.

  ```
  for i in range(1, 6):
      print(i)
  ```

- **while loop:** Runs until condition is false.

  ```
  i = 1
  while i <= 5:
      print(i)
      i += 1
  ```

8. Differentiate between break, continue, and pass statements with examples.

Answer:

- **break** → exits loop immediately.

  ```
  for i in range(5):
      if i == 3:
          break
  print(i, end=" ")
  ```

Output: 0 1 2

- **continue** → skips current iteration.

  ```
  for i in range(5):
      if i == 3:
          continue
      print(i, end=" ")
  ```

Output: 0 1 2 4

- **pass** → does nothing (placeholder).

  ```
  for i in range(5):
      if i == 2:
          pass
      print(i, end=" ")
  ```

Output: 0 1 2 3 4

9. Explain Python operators with examples.

Answer: Elaborate them with examples

- **Arithmetic:** `+ - * / % // **`
- **Relational:** `> < >= <= == !=`
- **Logical:** `and, or, not`
- **Assignment:** `=, +=, -=`

- **Membership:** `in, not in`
- **Identity:** `is, is not`
- **Bitwise:** `&, |, ^, ~, <<, >>`

10. Write a program to print a multiplication table of a given number using loops.

```python
num = int(input("Enter a number: "))
for i in range(1, 11):
    print(f"{num} x {i} = {num*i}")
```

```
Enter a number: 3
3 x 1 = 3
3 x 2 = 6
3 x 3 = 9
3 x 4 = 12
3 x 5 = 15
3 x 6 = 18
3 x 7 = 21
3 x 8 = 24
3 x 9 = 27
3 x 10 = 30
```

11. What is the use of “else” clause for loops. Explain with examples.

A) The **`else` clause with loops** in Python is a special feature that runs **only when the loop finishes normally** (i.e., *not interrupted by a `break` statement*)

```
for i in range(5):
   if i == 3:
        break
   print(i, end=” ”)
else:
  print(“Loop finished successfully”)
```

Output: 0 1 2

- **continue** → skips current iteration.

```
for i in range(5):
    if i == 3:
        continue
```

```
        print(i, end=" ")
    else:
        print("Loop finished successfully")
```

Output: 0   1   2 4
Loop finished successfully

More 5-Mark Questions:

Operators

11. With examples, illustrate the use of assignment and augmented assignment operators.
A)

```
x = 5 # assignment
x += 3 # augmented assignment (x becomes 8)
x , y = 4, 5 # Multiple assignment
```

12. Explain membership operators (`in`, `not in`) with examples.

```
X = [3, 5, 7, 6, 2, 4]
print(3 in x)  # Output: True
print(6 not in x)  # Output: True
```

13. Differentiate between identity operators (`is`, `is not`) and relational operators with examples.

```
x = [1, 2, 3]
y = [1, 2, 3]

print(x == y)   # True → contents are the same
print(x is y)   # False → stored at different memory locations
print(x is not y)  # True
```

14. Explain logical operators in Python with examples.

```
a, b = True, False
print("a and b:", a and b)   # False
print("a or b:", a or b)     # True
print("not a:", not a)       # False
```

15. Explain Python's arithmetic operators with examples.

```
x = -10
y = 4
result = x + y * 2
print(result)  # -2
```

16. Discuss relational operators in Python with suitable programs.

```
a, b = 10, 20
print(a > b)     # False
print(a < b)     # True
print(a == b)   # False
print(a != b)  # True
```

Control Structures (Decision Making)

17. Write a Python program using “if statement” to check whether a number is positive.

```
a = 10
if (a >= 0) :
    print(“The no is positive”)
else:
    print(“The no is negative”)
```

Output: The no is positive

18. Write a program using if-elif-else to calculate grade of a student based on marks.

```
marks = int(input("Enter your marks (0-100): "))
if marks >= 90:
    grade = "A+"
elif marks >= 80:
    grade = "A"
elif marks >= 70:
    grade = "B"
elif marks >= 60:
    grade = "C"
elif marks >= 50:
    grade = "D"
elif marks >= 40:
    grade = "E"
else:
    grade = "F (Fail)"
print("Your grade is:", grade)
```

```
Enter your marks (0-100): 68
Your grade is: C
```

19. **Write a program using nested conditions to check whether a year is a leap year or not.**

```
year = int(input("Enter a year: "))

if year % 4 == 0:           # First check: divisible by 4
    if year % 100 == 0:     # If divisible by 100, then check further
        if year % 400 == 0:   # If divisible by 400 -> Leap Year
            print(year, "is a Leap Year")
        else:
            print(year, "is NOT a Leap Year")
    else:
        print(year, "is a Leap Year")
else:
    print(year, "is NOT a Leap Year")

```

```
Enter a year: 2016
2016 is a Leap Year
```

Looping

20. **Program to print the Fibonacci sequence using for and while loop.**

```
n = int(input("Enter n: "))
a, b = 0, 1
print("Fibonacci Series:", end=" ")
for i in range(n):
    print(a, end=" ")
    a, b = b, a+b
```

```
Enter N: 8
Fibonacci Series: 0 1 1 2 3 5 8 13
```

```
n = int(input("Enter n: "))
a, b = 0, 1
count = 0
while count < n:
    print(a, end=" ")
    a, b = b, a + b
    count += 1
```

```
Enter n: 6
```

21. **Write a program to calculate the factorial of a number using a **while loop**.**

```
num = int(input("Enter a number: "))
fact = 1
i = 1
while i <= num:
    fact *= i
    i += 1
print("Factorial is:", fact)
```

```
Enter a number: 6
Factorial is: 720
```

22. **Write a program to find GCD of the given numbers.**

```
a = 64
b = 4
print(f"The gcd of {a} and {b} is", end=" ")
while b:
    a, b = b, a % b
print(a)
```

```
The gcd of 64 and 4 is 4
```